

MULTI-FUNCTION ROBOT CAR USING **ARDUINO UNO**

A Final Year Project Report

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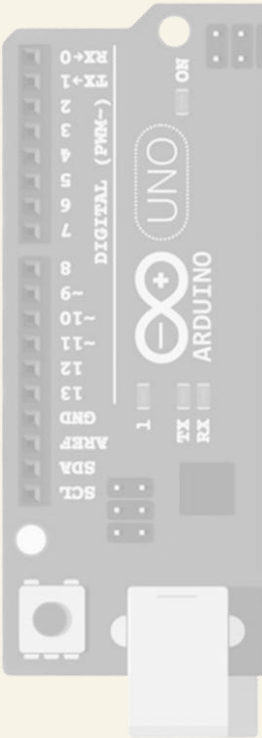
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Objectives of the Project

Design, build and program a multi-function robot car with 3 control modes in 1 platform

Obj. 1 – Chassis Construction

Build stable 4-wheel foam board chassis. Mount all electronics securely.

Obj. 2 – Obstacle Avoidance

HC-SR04 detects 12 cm or less-Stop, Backward, Scan, Turn to clear path.

Obj. 3 – Bluetooth Manual Control

Commands: F=Forward, B=Backward, L=Left, R=Right, S=Stop via smartphone app.

Obj. 4 – Voice Control with Safety

Voice commands via app. Safety check before Left or Right turn blocks if less than 10 cm.

Obj. 5 – Modular Code Design

3 separate functions: Obstacle, Bluetooth control, voice control 1-line mode switching.

Introduction

In today's era of automation, this project presents a **three-mode intelligent robot car** built on Arduino UNO combining autonomous sensing, wireless control, and voice interaction in one affordable platform.

Mode 2 - Bluetooth Control

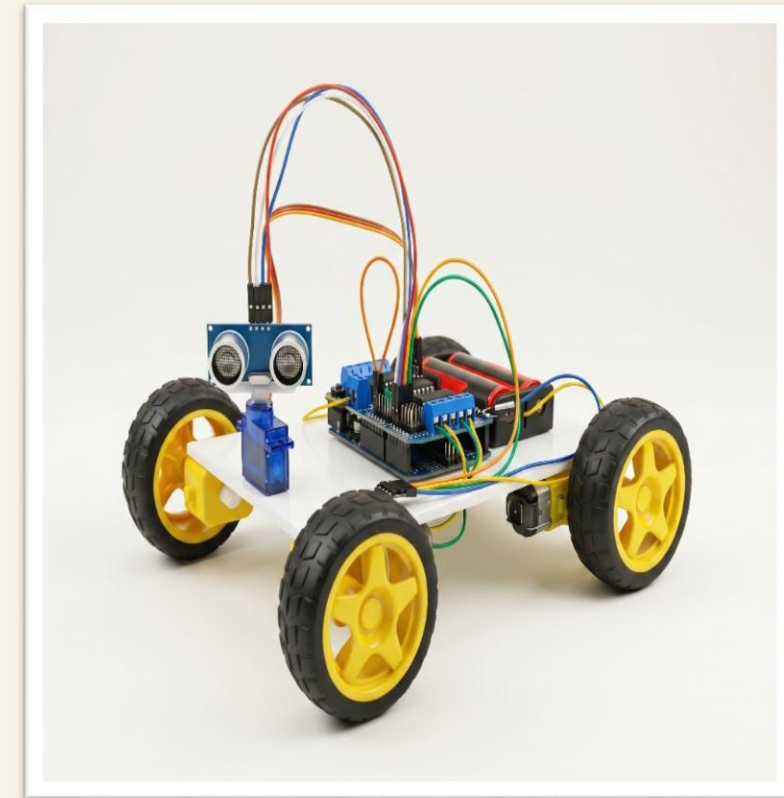
Autonomous navigation using HC-SR04 ultrasonic sensor.

Mode 2 - Bluetooth Control

Autonomous navigation using HC-SR04 ultrasonic sensor.

Mode 2 - Bluetooth Control

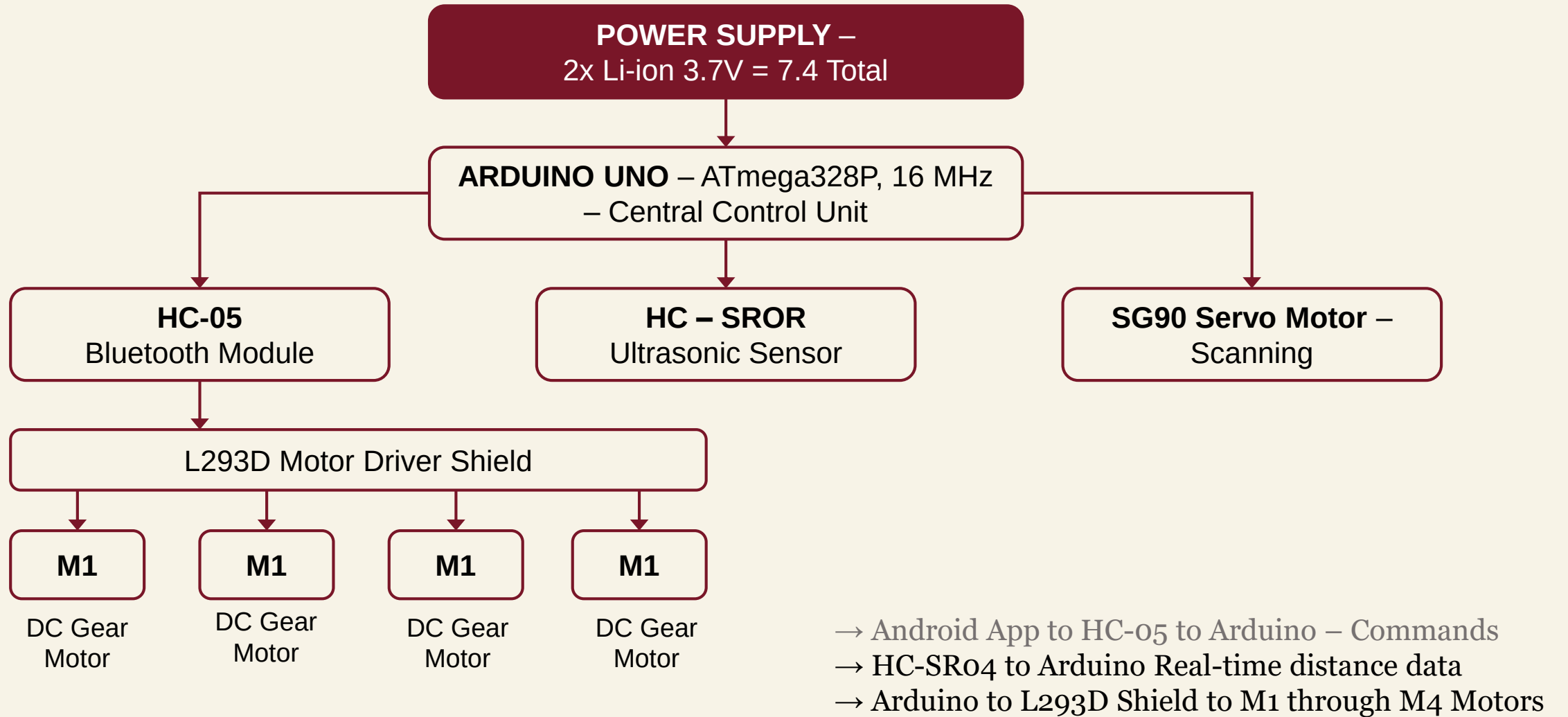
Autonomous navigation using HC-SR04 ultrasonic sensor.



Problem Addressed:

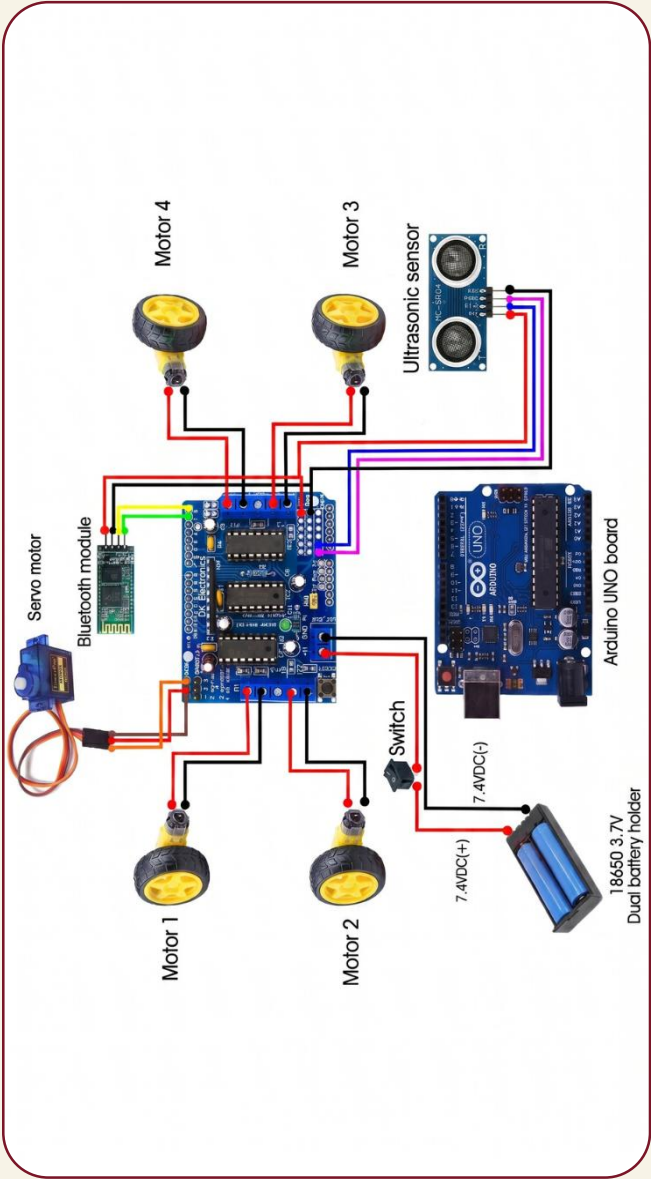
- × Low-cost robots support only 1 mode
- × Bluetooth cars execute commands blindly - no safety
- × Voice control rarely implemented at student level
- ✓ This project solves all three at approximately Rs.1,200.

System Block Diagram



Circuit Connection & Pin Configuration

Component	Pin	Connected To
HC-SR04 Ultrasonic Sensor	TRIG	Arduino A1 Digital Output
	ECHO	Arduino A0 Digital Input
	VCC	5V Power
	GND	GND Ground
SC90 Servo Motor	Signal (Orange)	Arduino Pin 10 PWM Output
	VCC (Red)	5V
	GND (Brown)	GND
HC-05 Bluetooth Module	TX	Arduino Pin 0 RX Serial In
	RX	Arduino Pin 1 TX Serial Out
	VCC	5V
	GND	GND
DC Motor M1 (Front-Left)	Both Wires	Shield M1 Terminal
DC Motor M2 (Rear-Left)	Both Wires	Shield M2 Terminal
DC Motor M3 (Front-Right)	Both Wires	Shield M3 Terminal
DC Motor M4 (Rear-Right)	Both Wires	Shield M4 Terminal
Battery Pack (7.4V)	Positive (+)	Shield EXT PWR positive
	Negative (-)	EXT PWR negative GND



Step-by-Step Working Principle

Mode 1 – Obstacle Avoidance

- Step 1:** Power ON - Servo centers at **103 degrees** - Motors speed set to **170**.
- Step 2:** HC-SR04 fires 10 microsecond pulse from TRIG pin A1.
- Step 3:** Distance equals Echo pulse divided by 29 divided by 2, result in cm.
- Step 4:** If distance greater than **12 cm** -execute forward function.
- Step 5:** If distance **12 cm** or less-Stop, Backward 100ms, Stop. Servo to 180 degrees, measure Left. Servo to 20 degrees, measure Right. If Left less than Right, turn right **500ms**. If Left greater than Right, turn left 500ms.

Mode 1 – Bluetooth Control

- Step 1:** HC-05 pairs with smartphone, PIN 1234.
- Step 2:** App sends character via UART at 9600 baud.
- Step 3:** Arduino reads via Serial.read function.

Character	Action
'F'	Forward
'B'	Backward
'L'	Left
'R'	Right
'S'	Stop



RoboNex



Bluetooth



Voice

Working Principle – Mode 3: Voice Control

Safety – Aware Voice Command System

Step 1: Voice command configured in app settings-5 total commands.

Step 2: User speaks, app recognizes speech, sends character via Bluetooth.

Step 3: Forward, Backward, Stop -immediate execution, no delay.

Step 4: Left-leftsee function checks side clearance first. If 10 cm or more, execute left turn 500ms then stop. If less than 10 cm, stop immediately - safety blocked.

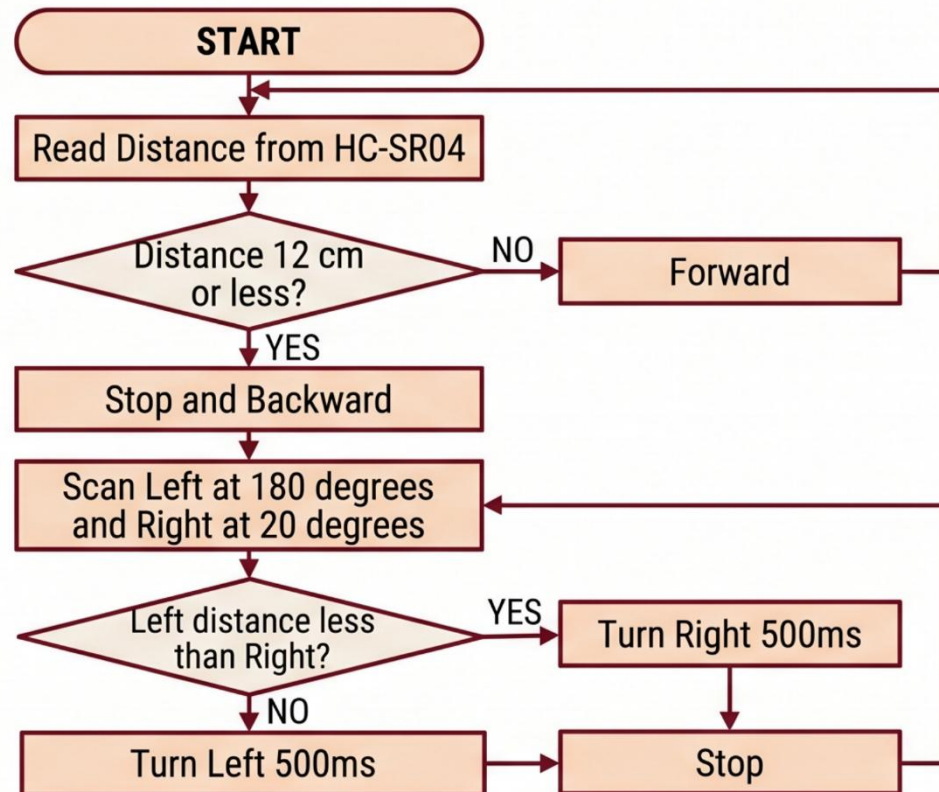
Step 5: Same safety logic applies for Right using rightsee function.

Voice Word	Character	Action
Forward	^	Move Forward
Backward	-	Move Backward
Left	<	Check and turn left
Right	>	Check and turn right
Stop	*	Stop all motors

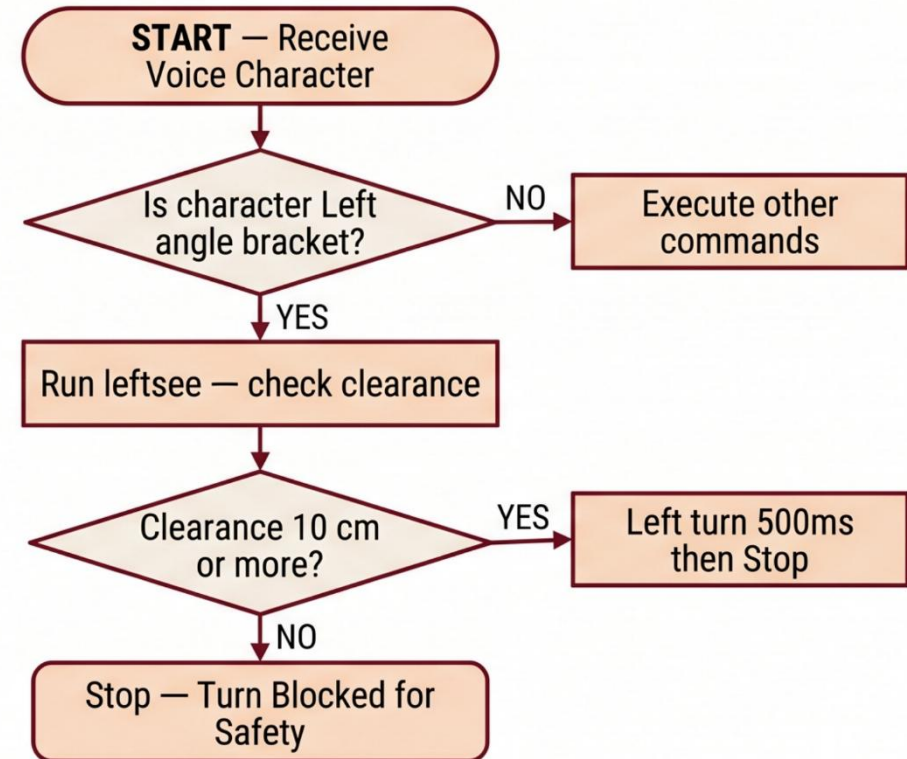
Safety Feature: If clearance is less than 10 cm on the turning side, the robot **stops automatically**. Collision prevention always active in Voice Mode.

System Flowchart

Flowchart 1 – Obstacle Avoidance Mode



Flowchart 2 – Voice Control



Software Platforms Used



Arduino IDE

Language: C and C++ Arduino flavored

Purpose: Write, compile and upload code

Feature: Serial Monitor for real-time debugging

Download: arduino.cc/en/software

Libraries Used

1. **Servo.h** -Controls SG90 servo motor signal, servo.write angle sets position
2. **AFMotor.h** - Controls L293D shield, FORWARD BACKWARD RELEASE and setSpeed 0 to 255



Bluetooth Controller App

Platform: Android smartphone

Purpose: Send directional and voice cmd's

Pairs with: HC-05, PIN 1234

Baud Rate: 9600 bps, D-pad and voice configuration

Mode Switching - one line at a tim

`void loop`

`// comment Obstacle Mode 1`

`// comment Bluetoothcontrol Mode 2`

`// comment voicecontrol Mode 3`

Hardware Components Used

No	Component	Specification	Qty
1	Arduino UNO R3	ATmega328P 16 MHz 32KB Flash	1
2	L293D Motor Shield Driver	4 Motors PWM 0 to 255 AFMotor Library	1
3	HC-SR04 Ultrasonic Sensor	Range 2 to 400 cm plus or minus 3mm accuracy	1
4	SG90 Servo Motor	0 to 180 degrees 1.8 kg cm torque	1
5	HC-05 Bluetooth Module	V2.0 approx. 10m range 2.4 GHz UART	1
6	DC Gear Motor	3 to 6V approx. 150 RPM 200mA	4
7	Robot Wheels	Compatible with gear motor shaft	4
8	Li-ion Battery 18650	3.7V 2600mAh each	2
9	Foam Board and Cardboard	Chassis structural material	-
10	Jumper Wires	Male to Male & Male to Female	set

Power Consumption Summary

Components	Current Draw
Arduino UNO	50 mA
4 Motors Loaded	800 mA
HC-SR04	15 mA
SG90 Servo	250 mA
HC-05	40 mA
	~ 1155 mA

Estimated Cost:

Rs. 800 to 1,200/-



Advantages & Disadvantages

Advantages

- ✓ **Multi-Modal:** 3 operating modes in 1 single platform.
- ✓ **Low Cost:** Complete build only Rs. 800 to Rs. 1200.
- ✓ **Safety Logic:** **Voice** mode blocks unsafe turning directions.
- ✓ **Modular Code:** Single line comment switches between modes.
- ✓ **No Router:** Direct Bluetooth pairing, no Wi-Fi needed.
- ✓ **Low Power-** HC-05 Bluetooth draws only approximately 40 mA.
- ✓ **Beginner Friendly:** Open-source Arduino platform.
- ✓ **Expandable:** Camera and GPS modules easily addable.
- ✓ **Educational:** Covers UART, PWM, ultrasonic sensing in one project.

Disadvantages

- ✗ **Limited Bluetooth Range:** Effective only up to approximately 10 metres.
- ✗ **Single Mode Only:** Cannot run two modes simultaneously.
- ✗ **No Camera:** No live video feed during operation
- ✗ **Fragile Chassis:** Foam board not waterproof or impact resistant
- ✗ **Sensor Limitation:** HC-SR04 struggles with soft or angled surfaces
- ✗ **No Battery Indicator:** Sudden shutdown without warning possible
- ✗ **iOS Incompatible:** HC-05 not supported on Apple devices
- ✗ **No Real-Time Feedback:** App does not display live distance or status

Test Results & Observation

Distance Accuracy Test

Actual cm	Measured cm	Error	Status
5 cm	5 cm	0 cm	STOP
10 cm	10 to 11 cm	± 1 cm	STOP
12 cm	12 to 13 cm	± 1 cm	Threshold
15 cm	15 cm	0 cm	FORWARD
30 cm	29 to 31 cm	± 1 cm	FORWARD

Bluetooth Response Test

- ✓ All 5 commands FBLRS response time less than **50 milliseconds**.
- ✓ Stable connection up to approximately **8 to 9 metres**.

Safety Check Test

- ✓ Obstacle at 6cm – Left Command – Robot **STOPPED**.
- ✓ Clear path at 25 cm – Right command – Robot **TURNED**.

Voice Command	Accuracy
FORWARD	90%
BACKWARD	80%
LEFT	90%
STOP	100%
Overall	88%

**5 out of 5
Objectives Achieved ✓**



Conclusion

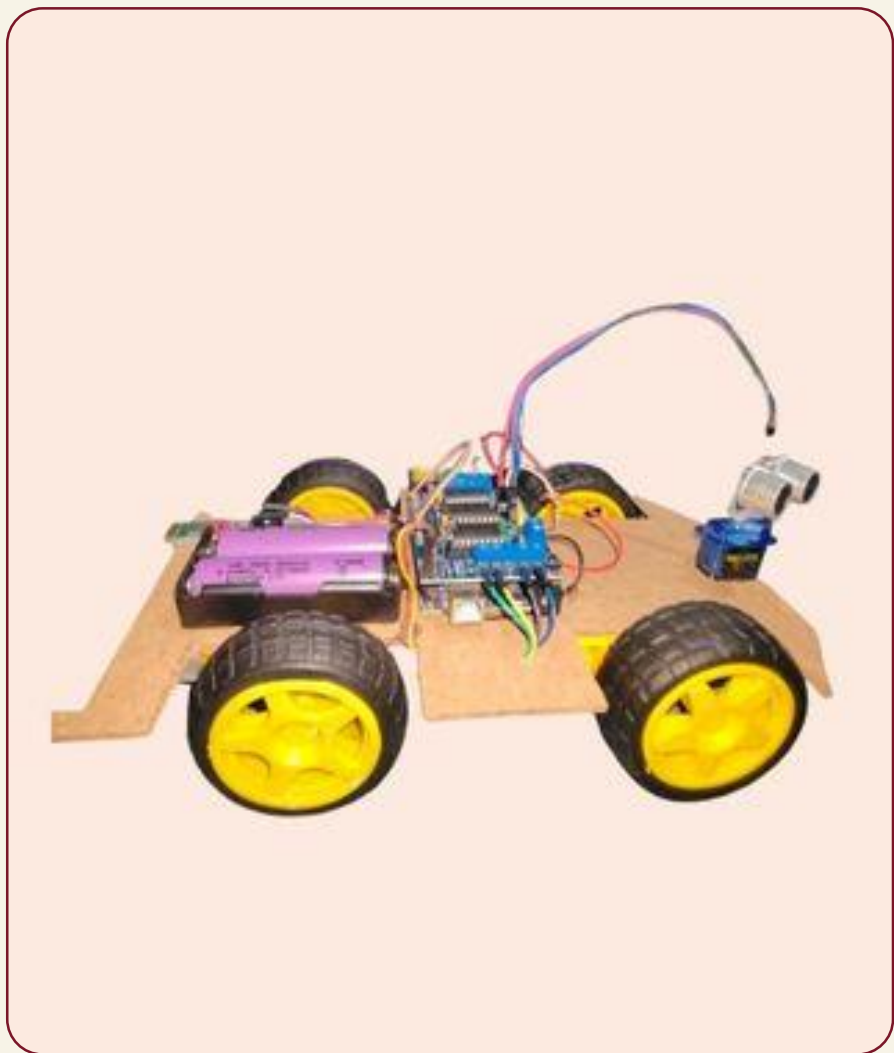
This project **successfully** designed and built a Multi-Function Robot Car using Arduino UNO-a cost-effective platform integrating **3 control modes** into a single embedded system.

Objective	Status	Key Result
Chassis Construction	✓ Done	Stable 4 wheel foam board chassis
Obstacle Avoidance	✓ Done	12 cm threshold ± 1 cm accuracy
Bluetooth Control	✓ Done	All 5 commands less than 50ms response
Voice and Safety	✓ Done	88% accuracy & 100% safe turn check
Modular Code	✓ Done	1 – line mode switching

Key Learnings

- UART serial communication HC-05 and Arduino
- PWM motor control via L293D shield
- Ultrasonic distance via pulseIn function
- Servo scanning for environment awareness
- Safety-aware human-robot voice interaction

Intelligent, multi-modal robots can be built at a fraction of commercial cost using open-source Arduino platform.



Future Scope & Enhancements

No	Enhancement	Technology	Benefit
1	Wi-Fi and IoT Control	ESP32 module	Control from anywhere via internet
2	Live Camera Feed	ESP32-CAM	Real-time video to smartphone
3	Custom Flutter App	Flutter Android and iOS	Joystick plus radar animation plus voice
4	360 Degree Scan	Stepper motor	Full-circle environment mapping
5	Battery Indicator	Voltage divider circuit	Prevents sudden shutdown
6	Speed Encoder	Hall effect sensor	Actual RPM closed-loop feedback
7	AI Object Detection	OpenCV with Raspberry Pi	Recognize specific objects autonomously
8	GPS Tracking	NEO-6M GPS module	Track robot location on map
9	Stronger Chassis	3D printed or acrylic	Waterproof and rigid structure

Immediate Next Steps

Add HC-05 Bluetooth status LED indicator
Speed control slider in Bluetooth app
Build custom RoboNex Flutter mobile app

Reference & Bibliography

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Datasheets

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2. **HC-05 AT Command Set** *Wavesen Technologies 2011*
3. **SG90 Servo Datasheet**, *TowerPro 2010*
4. Bluetooth Core Specification v2.0 plus EDR Bluetooth SIG with URL bluetooth.com

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THANK YOU

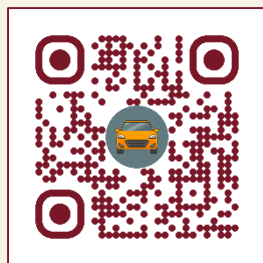
Thank you for your valuable time and attention.

We are grateful to:

Ms. Chinmayee Medhi - Project Supervisor and Guide

Mr. Amit Kumar Das - Head of Department

Principal Jitu Moni Deka - Chirang Polytechnic Bijni Assam



SCAN QR CODE FOR:

Complete Project Overview | Circuit Diagrams & Code |
FAQs & Troubleshooting | Video Demo & Working |
Future Enhancements

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